

3 (a) Rate of change of angular displacement [B1]

(b) (i) By conservation of energy,

Loss in E_p = Gain in E_k

$$mgh = \frac{1}{2}mv^2 - 0 \quad [\text{C1}]$$

$$m(9.81)(4.5 + 5.5) = \frac{1}{2}mv^2 \quad [\text{M1}]$$

$$v = 14 \text{ m s}^{-1} \quad [\text{A0}]$$

Do not accept use of kinematics equation.

$$\text{(ii)} \quad \omega = \frac{v}{r} = \frac{14}{4.5}$$

$$= 3.1 \text{ rad s}^{-1} \quad [\text{B1}]$$

(iii) (Tension – weight) provides for the centripetal force. [B1]

$$F_{\text{net}} = ma$$

$$T - mg = m \frac{v^2}{r}$$

$$T - (65)(9.81) = (65) \left(\frac{14^2}{4.5} \right) \quad [\text{C1}]$$

$$T = 3469 = 3470 \text{ N} \quad [\text{A1}]$$

(iv) Parabolic path with initial horizontal velocity [B1]